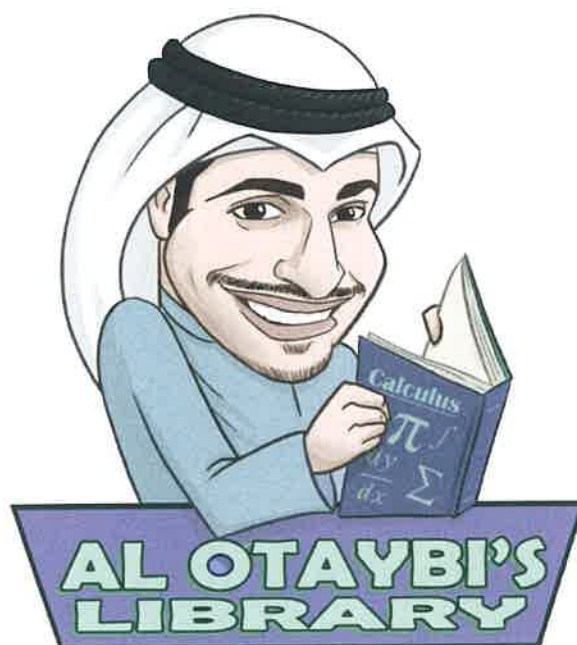


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 3.10 – 3.11 – 4.1

3.10 Linear Approximation and Differentials.

3.11 Hyperbolic Functions.

4.1 Maximum and Minimum Values.

(13)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



3.10 Linear approximation



طريقة تستخدم لتقريب ناتج عملية حسابية معقدة



شروط الحل



دالة $f(x)$

بداية x_0

نهاية x_1



$$\Delta x = x_1 - x_0$$

$$\frac{\pi}{180}$$

بالضرب في

degree

Rad.

إذا كانت بالـ

يتم تحويلها الى



خطوات الحل



1) f (نقطة البداية)

2) $f'(x)$

3) f' (نقطة البداية)

ثم يتم تطبيق القانون

$$f(x_1) \approx f(x_0) + f'(x_0)\Delta x$$

Use linear approximation to estimate $f(b)$ if the variable changes from a to b

$$f(x) = x^4, \quad a = 1, \quad b = 0.98$$

$$f(x) = x^4$$

$$x_0 = 1$$

$$x_1 = 0.98$$

$$\begin{aligned} \Delta x &= x_1 - x_0 \\ &= 0.98 - 1 \\ &= -0.02 \end{aligned}$$

$$f(x_0) = (1)^4 = 1$$

$$f'(x) = 4x^3$$

$$f'(x_0) = 4(1)^3 = 4(1) = 4$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$(0.98)^4 \approx 1 + 4(-0.02)$$

$$(0.98)^4 \approx 1 - 0.08$$

$$(0.98)^4 \approx 0.92$$

Use linear approximation to estimate $f(b)$ if the variable changes from a to b

$$f(x) = \sec x, \quad a = 60^\circ, \quad b = 62^\circ$$

$$f(x) = \sec x$$

$$x_0 = 60^\circ$$

$$x_1 = 62^\circ$$

$$\begin{aligned} \Delta x &= x_1 - x_0 \\ &= 62^\circ - 60^\circ \\ &= 2^\circ \left(\frac{\pi}{180} \right) \\ &= \frac{\pi}{90} \end{aligned}$$

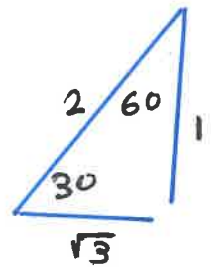
تم القرب بها
لأنها بار
degree

$$f(x_0) = \sec 60^\circ = \frac{1}{\cos 60^\circ} = \frac{1}{\frac{1}{2}} = 2$$

$$f'(x) = \sec x \tan x$$

$$f'(x_0) = \sec 60^\circ \tan 60^\circ$$

$$= 2 \cdot \frac{\sqrt{3}}{1} = 2\sqrt{3}$$



$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\sec 62^\circ \approx 2 + 2\sqrt{3} \left(\frac{\pi}{90} \right)$$

$$\approx 2 + \sqrt{3} \frac{\pi}{45}$$

$$\approx \frac{90}{45} + \frac{\sqrt{3} \pi}{45}$$

$$\approx \frac{90 + \sqrt{3} \pi}{45}$$

Use linear approximation to estimate

$$3(1.001)^3 + 5(1.001) + 7$$

$$f(x) = 3x^3 + 5x + 7$$

$$x_0 = 1$$

$$x_1 = 1.001 \longrightarrow \Delta x = x_1 - x_0$$

$$= 1.001 - 1 = 0.001$$

$$\begin{aligned} f(x_0) &= 3(1)^3 + 5(1) + 7 \\ &= 3 + 5 + 7 = 15 \end{aligned}$$

$$f'(x) = 9x^2 + 5$$

$$f'(x_0) = 9(1)^2 + 5 = 9 + 5 = 14$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$3(1.001)^3 + 5(1.001) + 7 \approx 15 + 14 \cdot (0.001)$$

$$\approx 15 + 0.014$$

$$\approx 15 + 0.014$$

$$\approx 15.014$$

Use linear approximation to estimate

$$\frac{1.01}{(1.01)^2 + 2}$$

$$f(x) = \frac{x}{x^2 + 2}$$

$$x_0 = 1$$

$$x_1 = 1.01$$

$$\Delta x = x_1 - x_0 = 1.01 - 1 = 0.01$$

$$f(x_0) = \frac{1}{(1)^2 + 2} = \frac{1}{1+2} = \frac{1}{3}$$

$$f'(x) = \frac{(x^2 + 2)(1) - x \cdot 2x}{(x^2 + 2)^2}$$

ماله داعي
لتبسيط

$$f'(x_0) = \frac{(1+2) - (1)2(1)}{(1+2)^3} = \frac{3-2}{(3)^2} = \frac{1}{9}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\frac{1.01}{(1.01)^2 + 2} \approx \frac{1}{3} + \frac{1}{9} (0.01)$$

$$\approx \frac{1}{3} + \frac{1}{9} \frac{1}{100}$$

$$\approx \frac{1}{3} + \frac{1}{900}$$

$$\approx \frac{300}{900} + \frac{1}{900}$$

$$\approx \frac{301}{900}$$

Use linear approximation to approximate

$$1 + (8.01)^{\frac{2}{3}}$$

$$f(x) = 1 + x^{\frac{2}{3}}$$

$$x_0 = 8$$

$$x_1 = 8.01 \longrightarrow \Delta x = x_1 - x_0 \\ = 8.01 - 8 \\ = 0.01$$

$$f(x_0) = 1 + (8)^{\frac{2}{3}} \\ = 1 + (2^3)^{\frac{2}{3}} = 1 + 4 = 5$$

$$f'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3x^{\frac{1}{3}}}$$

$$f'(x_0) = \frac{2}{3(8)^{\frac{1}{3}}} = \frac{2}{3(2)} = \frac{1}{3}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$1 + (8.01)^{\frac{2}{3}} \approx 5 + \frac{1}{3}(0.01)$$

$$\approx 5 + \frac{1}{3} \cdot \frac{1}{100}$$

$$\approx 5 + \frac{1}{300}$$

$$\approx \frac{1500}{300} + \frac{1}{300}$$

$$\approx \frac{1501}{300}$$

1, -2

0.998

9) Use differentials to estimate the number

$$N = \frac{\sqrt{2 - (1.001)^2}}{1.001}$$

$$f(x) = \frac{\sqrt{2 - x^2}}{x}$$

$$x_0 = 1 \rightarrow \Delta x = 1.001 - 1 = 0.001$$

$$x_1 = 1.001$$

$$1) f(x_0) = \frac{\sqrt{2-1}}{1} = \frac{1}{1} = 1$$

$$2) f'(x) = \frac{x \cdot \frac{-2x}{2\sqrt{2-x^2}} - \sqrt{2-x^2}}{x^2} \quad (1) \quad \text{ماده رايي}$$

$$3) f'(x_0) = \frac{(1) \cdot \frac{-1}{\sqrt{2-1}} - \sqrt{2-1}}{(1)^2}$$

$$= \frac{\frac{-1}{\sqrt{1}} - \sqrt{1}}{1} = -1 - 1 = -2$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 1 + (-2)(0.001)$$

$$\approx 1 - 0.002$$

$$\approx 0.998$$

Use linear approximation to estimate

$$2e^{0.1} + (0.1)^2$$

Apr 2019

$$f(x) = 2e^x + x^2$$

$$x_0 = 0$$

$$x_1 = 0.1$$

$$\longrightarrow \Delta x = x_1 - x_0 \\ = 0.1 - 0 = 0.1$$

$$f(x_0) = 2e^0 + 0 = 2(1) = 2$$

$$f'(x) = 2e^x + 2x$$

$$f'(x_0) = 2e^0 + 0 = 2(1) = 2$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 2 + 2(0.1)$$

$$\approx 2 + 0.2$$

$$\approx 2.2$$

Use linear approximation to estimate

Dec 2017

$$f(2.1), \text{ where } f(x) = \sqrt{x^3 + 1}$$

$$f(x) = \sqrt{x^3 + 1}$$

$$x_0 = 2$$

$$x_1 = 2.1$$

$$\longrightarrow \Delta x = x_1 - x_0 \\ = 2.1 - 2 = 0.1$$

$$f(x_0) = \sqrt{8 + 1} = \sqrt{9} = 3$$

$$f'(x) = \frac{3x^2}{2\sqrt{x^3 + 1}}$$

$$f'(x_0) = \frac{3(4)}{2\sqrt{8+1}} = \frac{3(4)}{2\sqrt{9}} = 2$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 3 + 2(0.1)$$

$$\approx 3 + 0.2$$

$$\approx 3.2$$

Let $f(x) = \tan^{-1}x - \frac{\pi}{4}$.

May 2017

Use linear approximation to estimate $f(1.05)$.

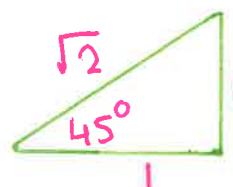
$$f(x) = \tan^{-1}x - \frac{\pi}{4}$$

$$x_0 = 1$$

$$x_1 = 1.05$$

$$\longrightarrow \Delta x = 1.05 - 1 = 0.05$$

$$\begin{aligned} f(x_0) &= \tan^{-1}1 - \frac{\pi}{4} \\ &= \frac{\pi}{4} - \frac{\pi}{4} = 0 \end{aligned}$$



$$f'(x) = \frac{1}{x^2 + 1}$$

$$f'(x_0) = \frac{1}{1 + 1} = \frac{1}{2}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 0 + \frac{1}{2} (0.05)$$

$$\approx \frac{1}{2} \quad \frac{\cancel{5}}{\cancel{100}_2}$$

$$\approx \frac{1}{2} \quad \frac{1}{20}$$

$$\approx \frac{1}{40}$$

Let $f(x) = \tan^{-1}x$

Use linear approximation to estimate $f(1.1)$

$$f(x) = \tan^{-1}x$$

$$x_0 = 1$$

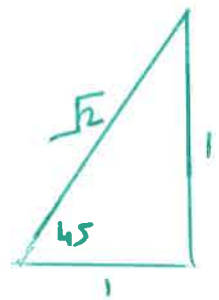
$$x_1 = 1.1$$

$$\longrightarrow \Delta x = x_1 - x_0 \\ = 1.1 - 1 = 0.1$$

$$f(x_0) = \tan^{-1}1 = \frac{\pi}{4}$$

$$f'(x) = \frac{1}{x^2 + 1}$$

$$f'(x_0) = \frac{1}{(1)^2 + 1} = \frac{1}{2}$$



$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\tan^{-1} 1.1 \approx \frac{\pi}{4} + \frac{1}{2} \cdot 0.1$$

$$\approx \frac{\pi}{4} + \frac{1}{2} \cdot \frac{1}{10}$$

$$\approx \frac{\pi}{4} + \frac{1}{20}$$

$$\approx \frac{5\pi}{20} + \frac{1}{20}$$

$$\approx \frac{5\pi + 1}{20}$$

Let $f(x) = 3x - \ln x$

Use linear approximation to estimate $f(1.1)$

$$f(x) = 3x - \ln x$$

$$x_0 = 1$$

$$x_1 = 1.1$$

$$\begin{aligned} \longrightarrow \Delta x &= x_1 - x_0 \\ &= 1.1 - 1 = 0.1 \end{aligned}$$

$$\begin{aligned} f(x_0) &= 3(1) - \ln 1 \\ &= 3 - 0 = 3 \end{aligned}$$

$$f'(x) = 3 - \frac{1}{x}$$

$$f'(x_0) = 3 - \frac{1}{1} = 3 - 1 = 2$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\begin{aligned} 3(1.1) - \ln(1.1) &\approx 3 + 2(0.1) \\ &\approx 3 + 0.2 \\ &\approx 3.2 \end{aligned}$$

Hw

Use Linear approximation to estimate

$$e^{0.1}$$

$$f(x) = e^x$$

$$x_0 = 0$$

$$x_1 = 0.1$$

$$\longrightarrow \Delta x = 0.1 - 0 = 0.1$$

انكسار

Use linear approximation to estimate

$$\sec^2(44.9^\circ)$$

$$f(x) = \sec^2 x = [\sec x]^2$$

$$x_0 = 45^\circ$$

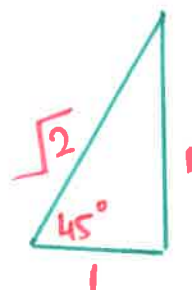
$$x_1 = 44.9^\circ$$

$$\begin{aligned} \longrightarrow \Delta x &= x_1 - x_0 \\ &= 44.9^\circ - 45^\circ \\ &= -0.1 \left(\frac{\pi}{180} \right) \end{aligned}$$

$$\begin{aligned} f(x_0) &= [\sec 45^\circ]^2 = \left[\frac{1}{\cos 45} \right]^2 \\ &= \left[\frac{1}{\frac{1}{\sqrt{2}}} \right]^2 = [\sqrt{2}]^2 = 2 \end{aligned}$$

$$\begin{aligned} f'(x) &= 2 [\sec x] \sec x \tan x \\ &= 2 \sec^2 x \tan x \end{aligned}$$

$$\begin{aligned} f'(x_0) &= 2 [\sec 45^\circ]^2 \tan 45^\circ \\ &= 2 \cdot \left[\frac{1}{\cos 45} \right]^2 (1) \\ &= 2 \left[\frac{1}{\frac{1}{\sqrt{2}}} \right]^2 = 2 [\sqrt{2}]^2 = 2 \cdot 2 = 4 \end{aligned}$$



$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\sec^2(44.9^\circ) \approx 2 + 4(-0.1) \left(\frac{\pi}{180} \right)$$

$$\approx 2 + 2 \left(-\frac{1}{10} \right) \left(\frac{\pi}{90} \right)$$

$$\approx 2 - \frac{1}{10} \left(\frac{\pi}{45} \right)$$

$$\approx 2 - \frac{\pi}{450}$$

$$\approx \frac{900}{450} - \frac{\pi}{450}$$

$$\approx \frac{900 - \pi}{450}$$

Let $f(x) = e^x \cos x$

Dec 2016

Use linear approximation to estimate $f(0.1)$.

$$f(x) = e^x \cos x$$

$$x_0 = 0$$

$$x_1 = 0.1$$

$$\longrightarrow \Delta x = x_1 - x_0 \\ = 0.1 - 0 = 0.1$$

$$f(x_0) = e^0 \cos 0 = (1)(1) = 1$$

$$f'(x) = e^x (1) \cos x + e^x (-\sin x)$$

اشتقاق بشتا + بشتا اشتقاق

$$f'(x_0) = e^0 \cos(0) + 0 = (1)(1) = 1$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 1 + 1(0.1)$$

$$\approx 1 + 0.1$$

$$\approx 1.1$$

Let $f(x) = \sin^2 \left(x^2 - 3x + \frac{\pi}{4} \right)$

(a) Find $f'(x)$

(b) Use a linear approximation to estimate $f(0.02)$

$$f(x) = \left[\sin(x^2 - 3x + \frac{\pi}{4}) \right]^2$$

a) $f'(x) = 2 \left[\sin(x^2 - 3x + \frac{\pi}{4}) \right] \cos(x^2 - 3x + \frac{\pi}{4}) [2x - 3]$

b) $f(x) = \left[\sin(x^2 - 3x + \frac{\pi}{4}) \right]^2$

$x_0 = 0$

$x_1 = 0.02 \rightarrow \Delta x = x_1 - x_0 = 0.02 - 0 = 0.02$

$f(x_0) = \left[\sin(0 - 0 + \frac{\pi}{4}) \right]^2 = \left[\sin \frac{\pi}{4} \right]^2 = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$

المستقة لى إبارهان a

$f'(x_0) = 2 \left[\sin(0 - 0 + \frac{\pi}{4}) \right] \cos(0 - 0 + \frac{\pi}{4}) [0 - 3]$
 $= 2 \sin \frac{\pi}{4} \cos \frac{\pi}{4} (-3)$
 $= 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot (-3) = \cancel{2} \cdot \frac{1}{2} (-3) = -3$

$\Rightarrow f(x_1) \approx f(x_0) + f'(x_0) \Delta x$

$\approx \frac{1}{2} - 3(0.02)$

$\approx 0.5 - 0.06$

≈ 0.44

so
 $\frac{6}{44}$



Text book question



Use a linear approximation (or differentials) to estimate the given number

$$\sqrt[3]{1001}$$

$$f(x) = \sqrt[3]{x} = x^{\frac{1}{3}}$$

$$x_0 = 1000$$

$$x_1 = 1001$$

$$\Delta x = 1001 - 1000 = 1$$

$$f(x_0) = \sqrt[3]{1000} = 10$$

$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}} = \frac{1}{3 x^{\frac{2}{3}}}$$

$$f'(x_0) = \frac{1}{3 (1000)^{\frac{2}{3}}} = \frac{1}{3 (10^3)^{\frac{2}{3}}} = \frac{1}{3(100)} = \frac{1}{300}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 10 + \frac{1}{300} (1)$$

$$\approx \frac{3000}{300} + \frac{1}{300}$$

$$\approx \frac{3001}{300}$$



مطالب إضافية



(Formula – expression) of

$$dy$$

$$dy = f'(x)dx$$



إذا عطاني الشروط



تطبيق الخطوات ثم

$$dy = f'(x_0)\Delta x$$

(Formula – expression) of

$$\Delta y$$

$$\Delta y = f(x + \Delta x) - f(x)$$

(Formula – expression) of

Linearization

$$L(x) = f(x_0) + f'(x_0)\Delta x$$





Text book question



Find the Linearization $L(x)$ of the function at a .

$$f(x) = x^3 - x^2 + 3, \quad a = -2$$

$$f(x) = x^3 - x^2 + 3$$

$$x_0 = -2$$

$$x_1 = x$$

$$\Delta x = x - (-2) = x + 2$$

$$f(x_0) = -8 - 4 + 3 = -9$$

$$f'(x) = 3x^2 - 2x$$

$$f'(x_0) = 3(4) + 4 = 12 + 4 = 16$$

$$L(x) = f(x_0) + f'(x_0) \Delta x$$

$$= -9 + 16(x + 2)$$

$$= -9 + 16x + 32$$

$$= 16x + 23$$



Text book question



Find the Linearization $L(x)$ of the function at a .

$$f(x) = \sin x, \quad a = \pi/6$$

$$f(x) = \sin x$$

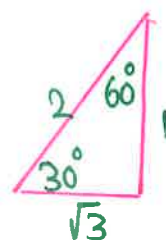
$$x_0 = \frac{\pi}{6}$$

$$x_1 = x \longrightarrow \Delta x = x - \frac{\pi}{6}$$

$$f(x_0) = \sin \frac{\pi}{6} = \frac{1}{2}$$

$$f'(x) = \cos x$$

$$f'(x_0) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$



$$\begin{aligned} L(x) &= f(x_0) + f'(x_0) \Delta x \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6} \right) \end{aligned}$$

Find the linearization of the function

$$f(x) = x \ln x \text{ at } a = 1$$

and use it to approximate $f(1.1)$.

July 2017

$$f(x) = x \ln x$$

$$x_0 = 1$$

$$x_1 = x$$

$$\longrightarrow \Delta x = x - 1$$

$$f(x_0) = (1) \ln 1 = 0$$

$$f'(x) = (1) \ln x + x \cdot \frac{1}{x}$$

$$f'(x_0) = \ln 1 + 1 = 0 + 1 = 1$$

$$L(x) = f(x_0) + f'(x_0) \Delta x$$

$$= 0 + 1 \cdot (x - 1)$$

$$= x - 1$$

$$f(1.1) \approx 1.1 - 1$$

$$\approx 0.1$$

Let $f(x) = \sqrt{x}$

(a) dy at $x = 9$ with $\Delta x = -0.1$

(b) Use differential to approximate $f(8.9)$

(a) $f(x) = \sqrt{x}$
 $x_0 = 9$
 x_1

$\Delta x = -0.1$

x_1 افترأ القاع
بسرعة
أمتابها

$f(x_0) = \sqrt{9} = 3$

$f'(x) = \frac{1}{2\sqrt{x}}$

$f'(x_0) = \frac{1}{2\sqrt{9}} = \frac{1}{2(3)} = \frac{1}{6}$

$dy = f'(x_0) \Delta x$

$= \frac{1}{6} \cdot (-0.1)$

$= \frac{1}{6} \left(-\frac{1}{10} \right) = -\frac{1}{60}$

(b) $f(x) = \sqrt{x}$
 $x_0 = 9$
 $x_1 = 8.9$
OK

$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$
 $\approx 3 + \left(-\frac{1}{60} \right)$ (د)

Let $y = \tan \theta$

- a) Use differentials to find dy if θ changes from 45° to 44°
b) Use a linear approximation to estimate $\tan 44^\circ$

$$y = \tan \theta$$

$$\theta_0 = 45^\circ$$

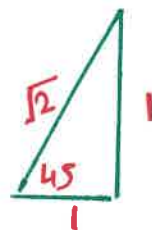
$$\theta_1 = 44^\circ$$

$$\longrightarrow \Delta \theta = 44^\circ - 45^\circ = -1^\circ \left(\frac{\pi}{180} \right) = -\frac{\pi}{180}$$

$$f(\theta_0) = \tan 45^\circ = 1$$

$$f'(\theta) = \sec^2 \theta$$

$$f'(\theta_0) = \sec^2 45 = \frac{1}{\cos^2 45} = \frac{1}{\left(\frac{\sqrt{2}}{2}\right)^2} = \frac{1}{\frac{1}{2}} = 2$$



$$(a) \quad dy = f'(\theta_0) \Delta \theta = 2 \left(-\frac{\pi}{180} \right) = -\frac{\pi}{90}$$

$$\theta_0 = 45^\circ$$

$$\theta_1 = 44^\circ$$

OK

$$(b) \quad \begin{aligned} f(\theta_1) &\approx f(\theta_0) + f'(\theta_0) \Delta \theta \\ &\approx 1 - \frac{\pi}{90} \\ &\approx \frac{90}{90} - \frac{\pi}{90} \\ &\approx \frac{90 - \pi}{90} \end{aligned}$$

Find the value of dy if $y = x^2 + \frac{1}{\pi} \sec^2\left(\frac{\pi}{12}x\right)$
and x changes from 3 to 3.03

$$y = x^2 + \frac{1}{\pi} \sec^2\left(\frac{\pi}{12}x\right)$$

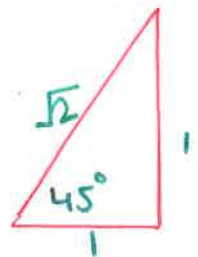
$$x_0 = 3$$

$$x_1 = 3.03$$

$$\begin{aligned} \Delta x &= x_1 - x_0 \\ &= 3.03 - 3 = 0.03 \end{aligned}$$

$$\begin{aligned} f(x_0) &= (3)^2 + \frac{1}{\pi} \sec^2\left(\frac{\pi}{12} \cdot 3\right) \\ &= 9 + \frac{1}{\pi} \sec^2\left(\frac{\pi}{4}\right) \\ &= 9 + \frac{1}{\pi} \frac{1}{\cos^2\left(\frac{\pi}{4}\right)} \\ &= 9 + \frac{1}{\pi} \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} \\ &= 9 + \frac{1}{\pi} \cdot 2 \end{aligned}$$

من إلقاء الطالب
المصورة
"المطلوب فقط dy"



$$\begin{aligned} f'(x) &= 2x + \frac{1}{\pi} \cdot 2 \sec\left(\frac{\pi}{12}x\right) \cdot \sec\left(\frac{\pi}{12}x\right) \tan\left(\frac{\pi}{12}x\right) \cdot \frac{\pi}{12} \\ &= 2x + \frac{1}{6} \left[\sec\left(\frac{\pi}{12}x\right) \right]^2 \tan\left(\frac{\pi}{12}x\right) \end{aligned}$$

$$\begin{aligned} f'(x_0) &= 2(3) + \frac{1}{6} \left[\sec\left(\frac{\pi}{12} \cdot 3\right) \right]^2 \tan\left(\frac{\pi}{12} \cdot 3\right) \\ &= 6 + \frac{1}{6} \left[\sec\left(\frac{\pi}{4}\right) \right]^2 \tan\left(\frac{\pi}{4}\right) \\ &= 6 + \frac{1}{6} \left[\frac{1}{\left(\cos \frac{\pi}{4}\right)^2} \right] (1) \\ &= 6 + \frac{1}{6} \left[\frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} \right] = 6 + \frac{1}{6} \cdot 2 \\ &= 6 + \frac{1}{3} \\ &= \frac{18}{3} + \frac{1}{3} = \frac{19}{3} \end{aligned}$$

$$\begin{aligned} \Rightarrow dy &= f'(x_0) \Delta x \\ &= \frac{19}{3} (0.03) \end{aligned}$$

$$= \frac{19}{3} \cdot \frac{3}{100} = \frac{19}{100} = 0.19$$

Differential dy

ما عطاني الشرح



تطبيق مباشر للقانون

$$dy = f'(x)dx$$

Example

Let $f(x) = x^2 - 5x$

Use differential to find the formula of dy

$$\begin{aligned} dy &= f'(x) dx \\ &= (2x - 5) dx \end{aligned}$$

عطاني الشرح



تطبيق الخطوات ثم

$$dy = f'(x_0)\Delta x$$

Example

Let $f(x) = x^2 - 5x$

Use differential to find dy if x changes from 1 to 1.1

$$f(x) = x^2 - 5x$$

$$x_0 = 1$$

$$x_1 = 1.1$$

$$\begin{aligned} \Delta x &= x_1 - x_0 \\ &= 1.1 - 1 \\ &= 0.1 \end{aligned}$$

$$f(x_0) = (1)^2 - 5(1) = -4$$

$$f'(x) = 2x - 5$$

$$f'(x_0) = 2(1) - 5 = -3$$

$$\begin{aligned} dy &= f'(x_0) \Delta x \\ &= (-3)(0.1) \\ &= -0.3 \end{aligned}$$



Text book questions



Find the differential of each function

$$y = xe^{-4x}$$

$$\begin{aligned} dy &= f'(x) dx \\ &= \left(\overset{\text{اشتقا}}{(1)} e^{-4x} + x \overset{\text{اشتقا}}{e^{-4x}} (-4) \right) dx \end{aligned}$$

Find the differential of each function

$$y = \ln(\sin \theta)$$

$$\begin{aligned} dy &= f'(\theta) d\theta \\ &= \frac{\cos \theta}{\sin \theta} d\theta \end{aligned}$$

a) Find the differential of each function

b) Evaluate dy for the given values of x and dx

$$y = \frac{x+1}{x-1}, \quad x = 2, \quad dx = 0.05$$

$$\begin{aligned} \text{a)} \quad dy &= f'(x) dx \\ &= \frac{(x-1)(1) - (x+1)(1)}{(x-1)^2} dx \end{aligned}$$

$$\begin{aligned} \text{b)} \quad \text{at } x=2, \quad dx=0.05 \\ dy &= \frac{(2-1) - (3)}{(2-1)^2} \cdot (0.05) \\ &= \frac{1-3}{(1)^2} (0.05) = (-2)(0.05) = -0.10 \end{aligned}$$

Let $f(x) = 1 + \sin x$

- (a) Use differentials to find the expression for Δy and dy
(b) Use linear approximation formula to estimate $f(29^\circ)$

$$\begin{aligned} \text{(a)} \quad \Delta y &= f(x + \Delta x) - f(x) \\ &= 1 + \sin(x + \Delta x) - [1 + \sin x] \\ &= \cancel{1} + \sin(x + \Delta x) - \cancel{1} - \sin x \\ &= \sin(x + \Delta x) - \sin x \end{aligned}$$

$$\begin{aligned} dy &= f'(x) dx \\ &= \cos x dx \end{aligned}$$

$$\text{(b)} \quad f(x) = 1 + \sin x$$

$$x_0 = 30^\circ$$

$$x_1 = 29^\circ$$

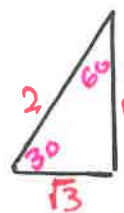
$$\begin{aligned} \Delta x &= x_1 - x_0 \\ &= 29^\circ - 30^\circ \\ &= -1^\circ \left(\frac{\pi}{180} \right) = -\frac{\pi}{180} \end{aligned}$$

$$f(x_0) = 1 + \sin 30^\circ = 1 + \frac{1}{2} = \frac{2}{2} + \frac{1}{2} = \frac{3}{2}$$

$$f'(x) = \cos x$$

$$f'(x_0) = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\begin{aligned} \Rightarrow f(x_1) &\approx f(x_0) + f'(x_0) \Delta x \\ &\approx \frac{3}{2} + \frac{\sqrt{3}}{2} \left(-\frac{\pi}{180} \right) \end{aligned}$$





Tricky question !!



Use linear approximation to show that if $x \approx 1$ then

$$\sqrt{x} \approx \frac{1}{2}(x+1)$$

$$f(x) = \sqrt{x}$$

$$x_0 = 1$$

$$x_1 = x$$

$$\longrightarrow \Delta x = x - 1$$

$$f(x_0) = \sqrt{1} = 1$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(x_0) = \frac{1}{2\sqrt{1}} = \frac{1}{2}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\sqrt{x} \approx 1 + \frac{1}{2}(x-1)$$

$$\sqrt{x} \approx 1 + \frac{1}{2}x - \frac{1}{2}$$

$$\sqrt{x} \approx \frac{2}{2} + \frac{1}{2}x - \frac{1}{2}$$

$$\sqrt{x} \approx \frac{1}{2} + \frac{1}{2}x$$

$$\sqrt{x} \approx \frac{1}{2}(1+x)$$



Tricky question !!



Use linear approximation to show that if $a = 0$ then

$$\sqrt[4]{1+2x} \approx 1 + \frac{1}{2}x$$

$$f(x) = \sqrt[4]{1+2x} = (1+2x)^{\frac{1}{4}}$$

$$x_0 = 0$$

$$x_1 = x$$

$$\longrightarrow \Delta x = x - 0 = x$$

$$f(x_0) = \sqrt[4]{1+0} = 1$$

$$f'(x) = \frac{1}{4} (1+2x)^{-\frac{3}{4}} \cdot 2 = \frac{2}{4(1+2x)^{\frac{3}{4}}} = \frac{1}{2(1+2x)^{\frac{3}{4}}}$$

$$f'(x_0) = \frac{1}{2(1+0)^{\frac{3}{4}}} = \frac{1}{2}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\sqrt[4]{1+2x} \approx 1 + \frac{1}{2}x$$

3.11 Hyperbolic Functions



Hyperbolic laws



$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\operatorname{csch} x = \frac{2}{e^x - e^{-x}}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\operatorname{coth} x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\operatorname{coth} x = \frac{\cosh x}{\sinh x}$$



Text book question



Find the numerical value of each expression

(1) $\sinh 0$

$$= \frac{e^0 - e^{-0}}{2}$$

$$= \frac{1 - 1}{2}$$

$$= 0$$

(2) $\tanh 0$

$$= \frac{e^0 - e^{-0}}{e^0 + e^{-0}}$$

$$= \frac{1 - 1}{1 + 1}$$

$$= \frac{0}{2} = 0$$

(3) $\cosh (\ln 5)$

$$= \frac{e^{\ln 5} + e^{-\ln 5}}{2}$$

$$= \frac{e^{\ln 5} + e^{\ln 5^{-1}}}{2}$$

$$= \frac{5 + \frac{1}{5}}{2}$$

$$= \frac{\frac{25}{5} + \frac{2}{5}}{2} = \frac{\frac{27}{5}}{2} = \frac{27}{10}$$

if $x = \ln\left(\frac{5}{2}\right)$, then $\cosh(x) =$

Nov 2018

- a) $\frac{29}{20}$ b) $\frac{21}{20}$ c) $\frac{20}{21}$ d) 0 e) None of the above

$$\cosh\left(\ln\frac{5}{2}\right)$$

$$\begin{aligned} &= \frac{e^{\ln\frac{5}{2}} + e^{-\ln\frac{5}{2}}}{2} \\ &= \frac{e^{\ln\frac{5}{2}} + e^{\ln\left(\frac{5}{2}\right)^{-1}}}{2} \\ &= \frac{\frac{5}{2} + \frac{2}{5}}{2} = \frac{\frac{25}{10} + \frac{4}{10}}{2} \\ &= \frac{\frac{29}{10}}{2} \\ &= \frac{29}{20} \quad (a) \checkmark \end{aligned}$$

The value of $\lim_{x \rightarrow \infty} \tanh x$ is

Apr 2019

a) does not exist. b) 0

c) $-\pi^2$ d) 1 e) None of the above

$$= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \frac{\infty - 0}{0 + \infty} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x \left[1 - \frac{e^{-x}}{e^x} \right]}{e^x \left[1 + \frac{e^{-x}}{e^x} \right]}$$

$$= \lim_{x \rightarrow \infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{1 - 0}{1 + 0} = 1$$

(d) ✓

* Remark *

On limit

$$e^{-\infty} = 0$$

$$e^{\infty} = \infty$$

The value of $\lim_{x \rightarrow \infty} \frac{\sinh x}{8e^x}$ is

Dec 2016

a) ∞ b) $\frac{1}{4}$ c) 0 d) $\frac{1}{16}$

e) None of the above

$$= \lim_{x \rightarrow \infty} \frac{\frac{e^x - e^{-x}}{2}}{8e^x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{16e^x} \quad \frac{\infty - 0}{\infty} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x \left[1 - \frac{e^{-x}}{e^x} \right]}{16e^x} = \lim_{x \rightarrow \infty} \frac{1 - e^{-2x}}{16} = \frac{1 - 0}{16} = \frac{1}{16}$$

(d) ✓



Text book questions



Prove the identity

$$\cosh x + \sinh x = e^x$$

$$\begin{aligned} \text{L.H.S} &= \cosh x + \sinh x \\ &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} + e^x - e^{-x}}{2} \\ &= \frac{2e^x}{2} = e^x = \text{R.H.S} \end{aligned}$$

Prove the identity

$$\cosh x - \sinh x = e^{-x}$$

$$\begin{aligned} \text{L.H.S} &= \frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} - (e^x - e^{-x})}{2} \\ &= \frac{e^x + e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^{-x}}{2} = e^{-x} = \text{R.H.S} \end{aligned}$$



Text book questions



Prove the identity

$$\sinh(-x) = -\sinh x$$

(This shows that $\sinh$ is an odd function)

$$\text{L.H.S} = \sinh(-x) = \frac{e^{(-x)} - e^{-(-x)}}{2} = \frac{e^{-x} - e^x}{2} \rightarrow (1)$$

$$\text{RHS} = -\sinh x = -\frac{e^x - e^{-x}}{2} = \frac{-e^x + e^{-x}}{2} = \frac{e^{-x} - e^x}{2} \rightarrow (2)$$

From (1) (2) $\rightarrow \therefore \text{L.H.S} = \text{R.H.S}$

Prove the identity

$$\cosh(-x) = \cosh x$$

(This shows that $\cosh$ is an even function)

$$\text{L.H.S} = \cosh(-x) = \frac{e^{(-x)} + e^{-(-x)}}{2} = \frac{e^{-x} + e^x}{2} \rightarrow (1)$$

$$\text{R.H.S} = \cosh(x) = \frac{e^x + e^{-x}}{2} \rightarrow (2)$$

From (1) (2) $\rightarrow \therefore \text{L.H.S} = \text{R.H.S}$



Text book question



Prove the identity

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\begin{aligned} \text{L.H.S} &= \sinh(x+y) \\ &= \frac{e^{(x+y)} - e^{-(x+y)}}{2} \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= \sinh x \cosh y + \cosh x \sinh y \\ &= \frac{e^x - e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2} \\ &= \frac{(e^x - e^{-x})(e^y + e^{-y})}{4} + \frac{(e^x + e^{-x})(e^y - e^{-y})}{4} \\ &= \frac{\cancel{e^{x+y}} + \cancel{e^{x-y}} - \cancel{e^0} - \cancel{e^{-x-y}}}{4} + \frac{\cancel{e^{x+y}} - \cancel{e^{x-y}} + \cancel{e^0} - \cancel{e^{-x-y}}}{4} \\ &= \frac{2e^{x+y} - 2e^{-x-y}}{4} \\ &= \frac{2(e^{x+y} - e^{-x-y})}{4} \\ &= \frac{e^{x+y} - e^{-(x+y)}}{2} = \text{L.H.S} \end{aligned}$$



Derivatives of Hyperbolic Functions



Function	Derivative
$f(x) = \sinh x$	$f'(x) = \cosh x$
$f(x) = \cosh x$	$f'(x) = \sinh x$
$f(x) = \tanh x$	$f'(x) = \operatorname{sech}^2 x$
$f(x) = \operatorname{csch} x$	$f'(x) = -\operatorname{csch} x \cdot \coth x$
$f(x) = \operatorname{sech} x$	$f'(x) = -\operatorname{sech} x \cdot \tanh x$
$f(x) = \coth x$	$f'(x) = -\operatorname{csch}^2 x$



Text book question



Find the derivative of

$$f(x) = \tanh \sqrt{x}$$

$$f'(x) = \operatorname{sech}^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$f(x) = \sinh(x^2)$$

$$f'(x) = \cosh(x^2) \cdot 2x$$

$$f(x) = \cosh(\ln x)$$

$$f'(x) = \sinh(\ln x) \cdot \frac{1}{x}$$

$$f(x) = e^{\cosh 3x}$$

$$f'(x) = e^{\cosh 3x} \cdot \sinh 3x \cdot (3)$$



Text book questions



Find the derivative of

$$g(x) = \sinh^2 x$$

$$g'(x) = 2 \sinh x \cosh x$$

$$f(x) = e^x \cosh x$$

اشتقاق بـ e^x + مشتق $\cosh x$

$$f'(x) = e^x (1) \cosh x + e^x \sinh x$$



Nice question



Find the derivative of

$$y = (\sinh x)^x$$

$$\ln y = \ln (\sinh x)^x$$

$$\ln y = x \ln (\sinh x)$$

Diff. w.r.t. x

اشتقاق بـ x + مشتق $\ln(\sinh x)$

$$\frac{y'}{y} = (1) \ln(\sinh x) + x \frac{\cosh x}{\sinh x}$$

$$y' = y [\ln \sinh x + x \coth x]$$

$$= (\sinh x)^x [\ln \sinh x + x \coth x]$$

Find the slope of the tangent line at the point $(0, 0)$ to the curve with equation

$$e^{xy} + \sin(x^2) = \cosh x + y$$

Dec 2017

Diff. w.r. t. x

$$e^{xy} \left[(1) y + x y' \right] + \cos(x^2) 2x = \sinh x + y'$$

at $(0, 0)$

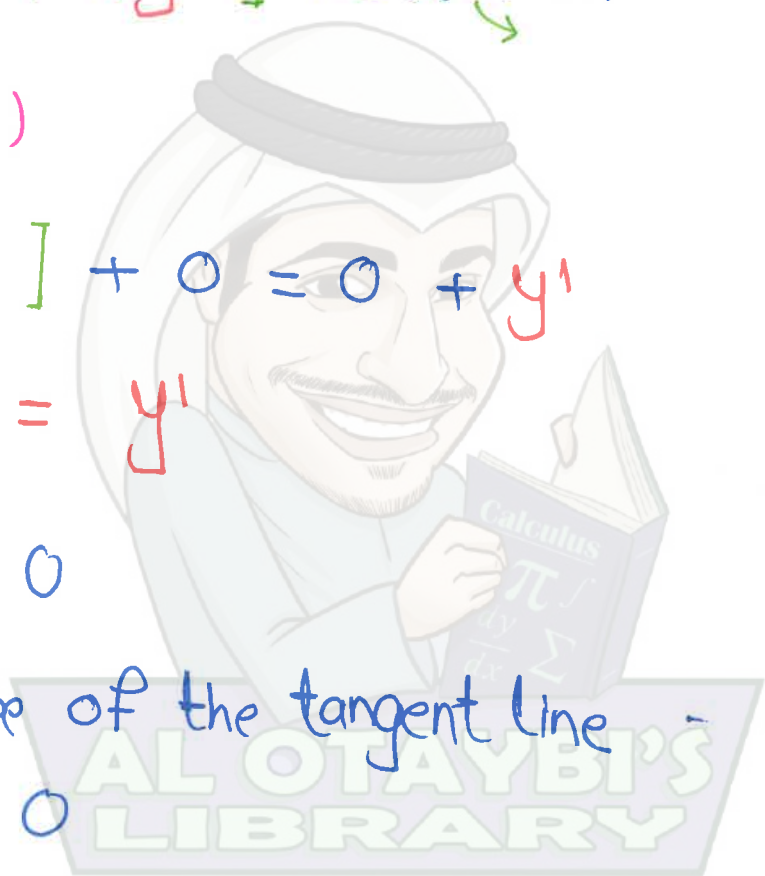
$$e^0 [0 + 0] + 0 = 0 + y'$$

$$(1) [0] = y'$$

$$y' = 0$$

∴ The slope of the tangent line

$$m = 0$$





زلقة بطيحة !!



$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

If $\cosh x = \frac{5}{3}$, $x > 0$.

Find the values of the other hyperbolic functions at x

$$\cosh^2 x - \sinh^2 x = 1$$

$$\left(\frac{5}{3}\right)^2 - \sinh^2 x = 1$$

$$\frac{25}{9} - 1 = \sinh^2 x$$

$$\sinh^2 x = \frac{25}{9} - \frac{9}{9}$$

$$\sinh^2 x = \frac{16}{9}$$

$$\sinh x = \frac{4}{3}$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{\frac{4}{3}}{\frac{5}{3}} = \frac{4}{5}$$

$$\sinh x = \frac{4}{3} \longrightarrow \operatorname{csch} x = \frac{3}{4}$$

$$\cosh x = \frac{5}{3} \longrightarrow \operatorname{sech} x = \frac{3}{5}$$

$$\tanh x = \frac{4}{5} \longrightarrow \coth x = \frac{5}{4}$$

The derivative of $\cosh^2 x - \sinh^2 x$ is

July 2017

- a) 0 b) $4\cosh x \sinh x$ c) $-2\cosh x \sinh x$
d) $2\cosh x \sinh x$ e) None of the above

$$\cosh^2 x - \sinh^2 x$$

Diff. w.r.t. x

$$2\cosh x \sinh x - 2\sinh x \cosh x = 0$$

(a) ✓

Solution #2

$$\cosh^2 x - \sinh^2 x = 1$$

Diff. w.r.t. x

$$(1)' = 0 \quad (a) \quad \checkmark$$

4.1 Maximum and Minimum Values



Critical numbers



1) Domain of $f(x)$



$$2) f'(x) \rightarrow$$

الممنوعات
(1) ممنوع أس سالب.
(2) ممنوع مقام غير موحد.

$= 0$

undefind

أصفار بسط المشتقة

أصفار مقام المشتقة

ثم يتم التأكد من انتمائها للمجال

الأعداد الناتجة يطلق عليها

critical numbers





Text book question



Find the critical number numbers of the function

$$F(x) = x^{4/5}(x-4)^2$$

Domain of $F(x) = \mathbb{R}$

$$F'(x) = \frac{4}{5} x^{-\frac{1}{5}} (x-4)^2 + x^{\frac{4}{5}} 2(x-4)$$

$$= \frac{4(x-4)^2}{5x^{\frac{1}{5}}} + x^{\frac{4}{5}} 2(x-4)$$

$$= \frac{4(x-4)^2}{5x^{\frac{1}{5}}} + \frac{5x^{\frac{1}{5}} x^{\frac{4}{5}} 2(x-4)}{5x^{\frac{1}{5}}}$$

$$= \frac{4(x-4)^2 + 10x(x-4)}{5x^{\frac{1}{5}}}$$

$$= \frac{(x-4)[4(x-4) + 10x]}{5x^{\frac{1}{5}}}$$

$$= \frac{(x-4)[4x-16+10x]}{5x^{\frac{1}{5}}}$$

$$= \frac{(x-4)[14x-16]}{5x^{\frac{1}{5}}}$$

$= 0$

$$(x-4)[14x-16] = 0$$

$$x-4 = 0$$

$$x = 4$$

$\in D_f$

$$14x-16 = 0$$

$$14x = 16$$

$$x = \frac{16}{14}$$

$$x = \frac{8}{7}$$

$\in D_f$

undefined

$$5x^{\frac{1}{5}} = 0$$

$\div 5$

$$x^{\frac{1}{5}} = 0$$

$$x = 0$$

$\in D_f$



Text book question



Find the critical numbers of the function

$$f(x) = x^2 e^{-3x}$$

Domain of $f(x) = \mathbb{R}$

$$f'(x) = 2x e^{-3x} + x^2 e^{-3x} (-3)$$

$$= 2x e^{-3x} - 3x^2 e^{-3x}$$

$$= 0$$

$$2x e^{-3x} - 3x^2 e^{-3x} = 0$$

$$x e^{-3x} [2 - 3x] = 0$$

$$x = 0$$

$\in D_f$

$$e^{-3x} = 0$$

$$2 - 3x = 0$$

$$2 = 3x$$

$$x = \frac{2}{3}$$

$\in D_f$

$\therefore$ The critical numbers are

$$x = 0 \quad x = \frac{2}{3}$$



Text book question



Find the critical numbers of the function

$$f(\theta) = 2\cos\theta + \sin^2\theta$$

Domain of $f(x) = \mathbb{R}$ [Trigonometric functions]

$$f'(\theta) = -2\sin\theta + 2\sin\theta \cos\theta$$

$$= 0 \quad \text{undefind}$$

$$-2\sin\theta + 2\sin\theta \cos\theta = 0 \quad \text{None}$$

$$2\sin\theta [-1 + \cos\theta] = 0$$

$$2\sin\theta = 0$$

$$\div 2$$

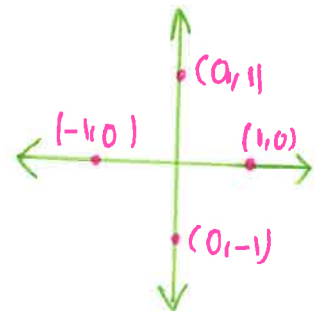
$$\sin\theta = 0$$

$$-1 + \cos\theta = 0$$

$$\cos\theta = 1$$

$$\theta = 0 + 2n\pi$$

$$\theta = 0 + n\pi$$



Where n is integer number.

Let $f(x) = \ln(x^2 - 1)$ Then :

- a) $x = 1$ is a critical number of f .
- b) $x = -1$ is a critical number of f .
- c) $x = 0$ is a critical number of f .
- d) $x = -2$ is a critical number of f .
- e) None of the above.

Nov 2017

Domain of $f(x)$: $x^2 - 1 > 0 \rightarrow x^2 > 1$
 $\Rightarrow D_f = \mathbb{R} \setminus [-1, 1]$

$$f'(x) = \frac{2x}{x^2 - 1}$$

$= 0$
 $2x = 0$

$x = 0 \notin D_f$

undefined

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$x = \pm 1 \notin D_f$

$\Rightarrow \therefore$ No critical point.

(e) ✓

Absolute Extrema on interval

((Absolute maximum – Absolute minimum))

$$1) f'(x) \rightarrow$$



الممنوعات

(1) ممنوع أس سالب.

(2) ممنوع مقام غير موحد.

$= 0$

أصفار بسط المشتقة

undefind

أصفار مقام المشتقة

ثم يتم التأكد من انتمائها للفترة

الأعداد الناتجة
 $f($ + $)$
أطراف الفترة



الناتج الأصغر هو

Absolute Minimum

الناتج الأكبر هو

Absolute Maximum





Text book question



Find the absolute maximum and absolute minimum values of f on the given interval

$$f(x) = 2x^3 - 3x^2 - 12x + 1, \quad [-2, 3]$$

$$f'(x) = 6x^2 - 6x - 12$$

$$= 0$$

$$6x^2 - 6x - 12 = 0$$

$$\div 6$$

$$x^2 - x - 2 = 0$$

$$(x + 1)(x - 2) = 0$$

$$x + 1 = 0$$

$$x = -1$$

$$\in [-2, 3]$$

undefind
none

$$x - 2 = 0$$

$$x = 2$$

$$\in [-2, 3]$$

$$\begin{aligned} f(-1) &= 2(-1) - 3(1) - 12(-1) + 1 \\ &= -2 - 3 + 12 + 1 = 8 \end{aligned}$$

$$\begin{aligned} f(2) &= 2(8) - 3(4) - 12(2) + 1 \\ &= 16 - 12 - 24 + 1 = -19 \end{aligned}$$

$$\begin{aligned} f(-2) &= 2(-8) - 3(4) - 12(-2) + 1 \\ &= -16 - 12 + 24 + 1 = -3 \end{aligned}$$

$$\begin{aligned} f(3) &= 2(27) - 3(9) - 12(3) + 1 \\ &= 54 - 27 - 36 + 1 = -8 \end{aligned}$$

$\therefore$ absolute maximum is 8 , absolute minimum is -19

Let the absolute maximum and minimum values of the function

$$f(x) = 3x - 5x^{3/5} \text{ on the interval } [-1, 32]$$

Dec 2016

$$f'(x) = 3 - 5 \cdot \frac{3}{5} x^{-2/5}$$

$$= 3 - \frac{3}{x^{2/5}}$$

$$= \frac{3x^{2/5}}{x^{2/5}} - \frac{3}{x^{2/5}}$$

$$= \frac{3x^{2/5} - 3}{x^{2/5}}$$

$= 0$

$$3x^{2/5} - 3 = 0$$

$$3x^{2/5} = 3$$

$\div 3$

$$x^{2/5} = 1$$

$$x = 1$$

$\in [-1, 32]$

undefined

$$x^{2/5} = 0$$

$$x = 0$$

$\in [-1, 32]$

$$f(1) = 3 - 5 = -2$$

$$f(0) = 0 - 0 = 0$$

$$f(-1) = -3 + 5 = 2$$

$$f(32) = 3(32) - 5(32)^{3/5}$$

$$= 96 - 5(2^3)^{3/5} = 96 - 5(8) = 96 - 40 = 56$$



$\therefore$ absolute maximum value is 56

$\therefore$ absolute minimum value is -2

Let $f(x) = x^3 - 3x + 1$

Find the absolute maximum and absolute minimum values of f on $[0, 3]$

$f(x) = x^3 - 3x + 1$ $[0, 3]$

Domain $\mathbb{R}$ $f(x) = 1\mathbb{R}$ [Polynomial of degree 3]
Cubic

$f'(x) = 3x^2 - 3$

$= 0$

$3x^2 - 3 = 0$

$3x^2 = 3$

$\div 3$

$x^2 = 1$

$x = -1$

$x = 1$

$\notin [0, 3]$

$\in [0, 3]$

$f(0) = 0 - 0 + 1 = 1$

$f(3) = 3^3 - 3(3) + 1 = 27 - 9 + 1 = 19$

$f(1) = 1 - 3 + 1 = -1$

$\Rightarrow$ the absolute maximum = 19

$\Rightarrow$ the absolute minimum = -1



Text book question



Find the absolute maximum and absolute minimum values of f on the given interval

$$f(x) = x + \frac{1}{x}, \quad [0.2, 4]$$

$$f'(x) = 1 - \frac{1}{x^2}$$

$$= \frac{x^2}{x^2} - \frac{1}{x^2}$$

$$= \frac{x^2 - 1}{x^2}$$

$$= 0$$

$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$x-1 = 0$$

$$x+1 = 0$$

$$x = 1$$

$$x = -1$$

$$\in [0.2, 4]$$

$$\notin [0.2, 4]$$

undefind

$$x^2 = 0$$

$$x = 0$$

$$\notin [0.2, 4]$$

$$f(1) = 1 + \frac{1}{1} = 1 + 1 = 2$$

$$f(0.2) = 0.2 + \frac{1}{0.2} = 0.2 + \frac{1}{\frac{2}{10}} = 0.2 + 5 = 5.2$$

$$f(4) = 4 + \frac{1}{4} = 4 + 0.25 = 4.25$$

∴ absolute maximum value is 5.2
absolute minimum value is 2

Let the absolute maximum and minimum values of

$$f(x) = x^2 e^{-2x} \text{ on the interval } [-1, 1]$$

May 2017

$$f'(x) = \overset{\text{اشتق}}{2x} \overset{\text{اشتق}}{e^{-2x}} + \overset{\text{اشتق}}{x^2} \overset{\text{اشتق}}{e^{-2x}} (-2)$$
$$= 2x e^{-2x} - 2x^2 e^{-2x}$$

$= 0$

undefined
None

$$2x e^{-2x} [1 - x] = 0$$

$$2x = 0$$

$$e^{-2x} = 0$$

$$1 - x = 0$$

$$x = 0$$

None

$$x = 1$$

$$\in [-1, 1]$$

$$\in [-1, 1]$$

$$f(0) = 0$$

$$f(1) = (1)^2 e^{-2} = \frac{1}{e^2}$$

$$f(-1) = (-1)^2 e^2 = e^2$$

$\therefore$ absolute maximum value is e^2

$\therefore$ absolute minimum value is 0



Text book question



Find the absolute maximum and absolute minimum values of f on the given interval

$$f(x) = \ln(x^2 + x + 1), \quad [-1, 1]$$

$$f'(x) = \frac{2x + 1}{x^2 + x + 1}$$

$= 0$

$$2x + 1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

$$\in [-1, 1]$$

undefind

$$x^2 + x + 1 = 0$$

$$a=1 \quad b=1 \quad c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

None

$$f\left(-\frac{1}{2}\right) = \ln\left(\frac{1}{4} - \frac{1}{2} + 1\right) = \ln\left(\frac{1}{4} - \frac{2}{4} + \frac{4}{4}\right) = \ln\left(\frac{3}{4}\right)$$

$$f(-1) = \ln(1 - 1 + 1) = \ln 1 = 0$$

$$f(1) = \ln(1 + 1 + 1) = \ln 3$$

∴ absolute maximum value is $\ln 3$

∴ absolute minimum value is $\ln\left(\frac{3}{4}\right)$

Find the absolute maximum and absolute minimum values , if

any , of $f(x) = \sin(2x) - 2x$ on the interval $[0, 2\pi]$

Nov 2018

$$f'(x) = \cos(2x)(2) - 2$$

$=0$ Undefined
none

$$\cos(2x)(2) - 2 = 0$$

$$2 \cos(2x) = 2$$

$\div 2$

$$\cos(2x) = 1$$

$$2x = 0$$

$$, 2x = 2\pi$$

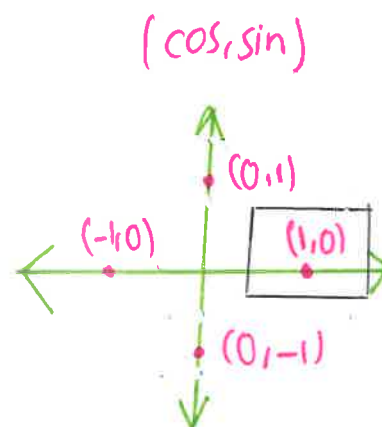
$\div 2$

$$x = 0$$

$$x = \pi$$

$$\in [0, 2\pi]$$

$$\in [0, 2\pi]$$



$$f(0) = 0 - 0 = 0$$

$$f(\pi) = \sin 2\pi - 2\pi = -2\pi$$

$$f(2\pi) = \sin 4\pi - 2(2\pi) = -4\pi$$

$\therefore$ absolute maximum value is 0

$\therefore$ absolute minimum value is -4π

Find the absolute maximum and absolute minimum values of

$$f(x) = \cos^2 x + \sin x \quad \text{on the interval } \left[0, \frac{\pi}{2}\right]$$

Apr 2019

$$\begin{aligned} f'(x) &= 2\cos x (-\sin x) + \cos x \\ &= -2\cos x \sin x + \cos x \end{aligned}$$

$= 0$

undefind

$$-2\cos x \sin x + \cos x = 0 \quad \text{None}$$

$$\cos x [-2\sin x + 1] = 0$$

$$\cos x = 0$$

$$-2\sin x + 1 = 0$$

$$1 = 2\sin x$$

$$\div 2$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6}$$

$$x \in \left[0, \frac{\pi}{2}\right]$$

$$f\left(\frac{\pi}{6}\right) = \cos^2 \frac{\pi}{6} + \sin \frac{\pi}{6} = \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

$$f(0) = \cos^2 0 + \sin 0 = 1 + 0 = 1$$

$$f\left(\frac{\pi}{2}\right) = 0 + \sin \frac{\pi}{2} = 1$$

∴ absolute maximum value is $\frac{5}{4}$
∴ absolute minimum value is 1

Find the absolute extrema of

$$f(x) = \sin x - \cos x \quad \text{on } [0, \pi]$$

$$f(x) = \sin x - \cos x \quad [0, \pi]$$

Domain of $f(x) = \mathbb{R}$ [Trigonometric function]

$$f'(x) = \cos x + \sin x$$

$$= 0$$

$$\cos x + \sin x = 0$$

$$\sin x = -\cos x$$

$$\div \cos x$$

$$\tan x = -1$$

$$\Rightarrow x = \frac{3\pi}{4} \in (0, \pi)$$

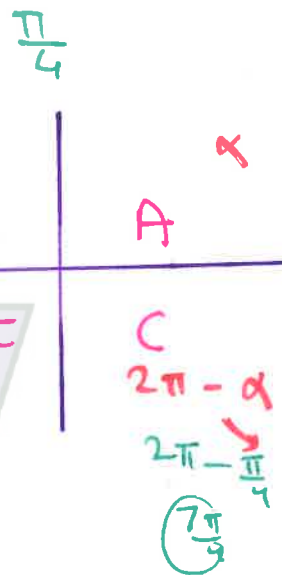
$$f(0) = \sin 0 - \cos 0 = -1$$

$$f(\pi) = \sin \pi - \cos \pi = 0 - (-1) = 1$$

$$f\left(\frac{3\pi}{4}\right) = \sin \frac{3\pi}{4} - \cos \frac{3\pi}{4} = \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}}$$

the absolute maximum is $\frac{2}{\sqrt{2}}$

the absolute minimum is -1



Let $f(x) = x^2 + 1$ on the interval $[-1, 1]$.

a) 1 is the absolute minimum value of f and 2 is the absolute maximum value of f .

b) $f(0.5)$ is a local minimum value of f .

c) f has no absolute extrema values.

d) $f(0.5)$ is a local maximum value of f .

e) None of the above.

Dec 2017

$$f'(x) = 2x$$

$$= 0$$

$$2x = 0$$

$$\div 2$$

$$x = 0 \in [-1, 1]$$

undefind
None

$$f(0) = 0 + 1 = 1 \rightarrow \text{abs. min. value} = 1$$

$$f(-1) = (-1)^2 + 1 = 1 + 1 = 2 \rightarrow \text{abs. max. value} = 2$$

$$f(1) = (1)^2 + 1 = 1 + 1 = 2$$

(a) ✓

(HW): Find the absolute extrema

$$f(x) = x \sqrt{4 - x^2} \text{ on } [-1, 2]$$

Ans abs. max. = 2

abs. min. = $-\sqrt{3}$

Let $f(x) = \ln(x^2 + 2x + 2)$ be defined on $[-2, 0]$.

One of the following statements is true

- a) 0 is the absolute maximum value of f .
- b) f has no critical numbers in $(-2, 0)$
- c) f has an absolute minimum value at $x = -1$.
- d) $x = -\ln 2$ is a critical number of f .
- e) None of the above.

Nov 2018

$$f'(x) = \frac{2x+2}{x^2+2x+2}$$

$= 0$ $\rightarrow$ undefind

$$2x+2=0$$
$$2x=-2$$
$$\div 2$$
$$x = -1 \in [-2, 0]$$
$$x^2+2x+2=0$$
$$a=1 \quad b=2 \quad c=2$$
$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$
$$= \frac{-2 \pm \sqrt{4-4(1)(2)}}{2(1)}$$
$$= \frac{-2 \pm \sqrt{-4}}{2}$$

None

$$f(-1) = \ln(1-2+2) = \ln 1 = 0$$

$$f(-2) = \ln(4-4+2) = \ln 2$$

$$f(0) = \ln(0+0+2) = \ln 2$$

(c) ✓



Tricky question !!



Find the largest slope of the tangent line of $f(x)$ on the given interval

$$f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2, \quad [-1, 1]$$

$$f'(x) = \frac{1}{3} 3x^2 - \frac{1}{2} 2x$$

$$= x^2 - x$$

$$\text{Let } h(x) = x^2 - x$$

$$h'(x) = 2x - 1$$

$$= 0$$

$$2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2} \in [-1, 1]$$

undefind
None

$$h\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{2} = \frac{1}{4} - \frac{2}{4} = \boxed{-\frac{1}{4}}$$

$$h(-1) = 1 - (-1) = 1 + 1 = \boxed{2}$$

$$h(1) = 1 - 1 = \boxed{0}$$

∴ The largest slope of tangent line = 2